

**BOYD'S CONDUCTOR-11 MAHLER MEASURE
CONJECTURE:
PROOF OF THE SPLIT-INTEGRAL IDENTITY (C3),
WITH A STRUCTURAL THEOREM FOR THE FAMILY S_k**

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ABSTRACT. Boyd's 1998 tables of conjectural identities $m(P) = r |L'(E, 0)|$ between Mahler measures of two-variable polynomials and L -values of elliptic curves begin with the smallest possible conductor, $N = 11$. Two of the three conductor-11 identities were proved by Brunault via an explicit version of Beilinson's theorem on modular units. The third one, (C3), concerns the polynomial $S_0 = y^2 + (x^2 + 1)y + x^3$, which vanishes on the unit torus, and asserts that a *signed split integral* I_{split} of $\log |y|$ around the branch cut equals $b_{11} = L'(E_{11}, 0)$. We prove (C3). The proof identifies the split integral with a regulator integral along Samart's signed open chain $\tilde{\gamma}$; we close $\tilde{\gamma}$ by a small-branch compensating arc β_0 into a closed anti-invariant integral cycle $C' = \tilde{\gamma} + \beta_0$, and prove that its homology class is $2\gamma^-$, where γ^- generates $H_1(E, \mathbb{Z})^-$: the period ratio $\text{period}(C')/w_{\text{anti}}$ is a-priori a non-zero integer, and a ball-arithmetic (Arb) computation with certified error bounds pins it to -2 . Combined with the exact integral identity $\int_{\beta_0} \eta = \int_{\tilde{\gamma}} \eta$ and the theorems of Bloch, Brunault and Bertin, this yields $I_{\text{split}} = b_{11}$, in agreement with the 149-digit numerical value. We also prove a structural theorem for the whole family $S_k = y^2 + (x^2 + kx + 1)y + x^3$: the dividing line is the torus-intersection structure, not the sign of k ; we confirm numerically (60 digits, with PARI/GP cross-checks) new Boyd-type identities for $k = 2, 3, -4, -5, -6$, and we adjudicate negatively the conductor-53 analogue suggested by Samart. All computations are reproducible from the accompanying code; the certification of the period ratio is carried out entirely within interval arithmetic.

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1. INTRODUCTION

1.1. Boyd’s conductor-11 question. Let $m(P)$ denote the logarithmic Mahler measure of a Laurent polynomial P . Boyd’s systematic numerical experiments [4] produced dozens of conjectural identities of the shape

$$m(P_k) = r_k |L'(E_k, 0)|, \quad r_k \in \mathbb{Q},$$

organised by the conductor N of the elliptic curve E_k cut out by $P_k = 0$. The smallest possible conductor of an elliptic curve over $\mathbb{Q}$ is $N = 11$ (there are no elliptic curves of conductor 1 through 10), so the conductor-11 rows head Boyd’s table. Three polynomials of conductor 11 appear:

$$(C1) \quad m((1+x)(1+y)(1+x+y)+xy) = 7b_{11},$$

$$(C2) \quad m(y^2 + (x^2 + 2x - 1)y + x^3) = 5b_{11},$$

$$(C3) \quad I_{\text{split}} = \pm b_{11},$$

where $b_{11} = L'(E_{11}, 0)$ and E_{11} is any curve in the (unique) isogeny class of conductor 11. Identities (C1) and (C2) were proved by Brunault [5] via his explicit version of Beilinson’s regulator theorem for modular units on $X_1(11)$.

The third case is different in kind. The polynomial

$$S_0 = y^2 + (x^2 + 1)y + x^3$$

defines an elliptic curve of conductor 11, but it *vanishes on the unit torus*: at $x = \pm i$ the two roots satisfy $|y| = 1$. The Mahler measure itself is *not* a rational multiple of b_{11}

$$m(S_0) = 0.40560295591501040\dots,$$

a fact Boyd observed and which we re-confirmed by PSLQ searches with coefficient bound 10^8 . Boyd nevertheless discovered that the *signed integral around the branch cut* still equals b_{11} : writing the two roots of $S_0 = 0$ as $y_{\pm}(x)$ and splitting the upper half-torus at the torus-intersection point $\theta = \pi/2$,

$$(C3) \quad I_{\text{split}} := \underbrace{\frac{1}{\pi} \int_0^{\pi/2} \log |y_-(e^{i\theta})| d\theta}_{I_1} - \underbrace{\frac{1}{\pi} \int_{\pi/2}^{\pi} \log |y_-(e^{i\theta})| d\theta}_{I_2} \stackrel{?}{=} \pm b_{11}.$$

Boyd's own words (quoted in [16]):

“This is in accord with our contention that in case P vanishes on the torus, it is the integral of $\log |y|$ around a branch cut rather than $m(P)$, which should be rationally related to $L'(E, 0)$.”

Samart [16] lists (C3) explicitly as an open conjecture.

1.2. Summary of results.

- (1) **Proof of (C3).** We prove $I_{\text{split}} = b_{11}$ (Theorem 4.3). The proof identifies Boyd's split-integral chain with a closed anti-invariant integral cycle $C' = \tilde{\gamma} + \beta_0$ on E whose homology class is $2\gamma^-$, where γ^- generates $H_1(E, \mathbb{Z})^-$; the integrality argument is made rigorous by a ball-arithmetic (Arb) computation of the period ratio with certified error bounds (§5.2). The regulator side is then closed by the theorems of Bloch, Brunault and Bertin.
- (2) **High-precision numerics.** (C3) is confirmed to 149 digits: $|I_{\text{split}} - b_{11}| = 4.85 \times 10^{-149}$ (previous public record: Boyd's 50 digits). (C1) and (C2) are independently reproduced to 52 digits.
- (3) **Structural identity.** On $[0, \pi]$ one has $|y_-(e^{i\theta})| \leq 1 \leq |y_+(e^{i\theta})|$, hence the exact identity $I_1 + I_2 = -m(S_0)$, which reduces (C3) to the individual values $I_1 = (b_{11} - m(S_0))/2$, $I_2 = -(b_{11} + m(S_0))/2$ (Proposition 3.1).
- (4) **A structural theorem for the family S_k .** For $S_k = y^2 + (x^2 + kx + 1)y + x^3$ the dividing line is the torus-intersection structure: for $|k| \geq 2$ the classical Deninger mechanism applies and $m(S_k) = r_k |L'(E_k, 0)|$; for $-4 < k < 2$ a Boyd-type identity holds if and only if the curve has torsion making x, y modular units, which happens only for $k = 0$ and $k = 1$ (Theorem 3.4).

- (5) **New numerically confirmed identities** (60 digits, PARI/GP cross-checked): $m(S_2) = 2|L'(E_{37}, 0)|$, $m(S_3) = |L'(E_{79}, 0)|$, and the predicted-then-confirmed identities $m(S_{-4}) = \frac{7}{2}|L'(E_{37}, 0)|$, $m(S_{-5}) = \frac{1}{4}|L'(E_{359}, 0)|$, $m(S_{-6}) = \frac{1}{8}|L'(E_{997}, 0)|$.
- (6) **Negative adjudication of the conductor-53 case.** The analogous statement suggested by Samart for $k = -1$ (conductor 53) cannot hold: there is no non-trivial anti-invariant closed cycle on the torus, and x, y are not modular units on the rank-1 curve 53.a1 (Theorem 3.6).

1.3. Organisation and reproducibility. Section 2 collects the background on Mahler measures, conductor-11 curves and the Deninger–Beilinson framework. Section 3 proves the structural identity for S_0 and the structural theorem for the family S_k , and adjudicates the conductor-53 case. Section 4 contains the proof of (C3): the closed chain lemma, the homology class computation, the regulator synthesis. Section 5 describes the numerical certification, including the ball-arithmetic proof of the period ratio (§5.2). Section 6 states the conjectures suggested by our computations. Appendix A lists the reproduction code. All scripts (Python/mpmath, PARI/GP, python-flint/Arb), raw outputs, and the sources of this paper are available in the accompanying repository.

2. BACKGROUND

2.1. Mahler measure. For a non-zero Laurent polynomial $P \in \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$, the *logarithmic Mahler measure* is the average of $\log |P|$ over the unit torus $\mathbb{T}^n = \{|x_1| = \dots = |x_n| = 1\}$:

$$m(P) = \int_0^1 \cdots \int_0^1 \log |P(e^{2\pi i t_1}, \dots, e^{2\pi i t_n})| dt_1 \cdots dt_n, \quad M(P) = e^{m(P)}.$$

In one variable Jensen’s formula gives, for $P(x) = a_0 \prod_{j=1}^d (x - \alpha_j)$,

$$m(P) = \log |a_0| + \sum_{j=1}^d \log^+ |\alpha_j|, \quad \log^+ v := \max(\log v, 0).$$

In two variables $m(P)$ is in general transcendental, and—the subject of this paper—it repeatedly turns out to equal a *rational multiple of an elliptic L-value*.

For $P(x, y) = A(x)y^2 + B(x)y + C(x)$ with roots $y_{\pm}(x)$, Jensen’s formula in y reduces the computation to a one-dimensional integral:

$$(1) \quad m(P) = \frac{1}{2\pi} \int_0^{2\pi} \left[\log |A(e^{i\theta})| + \log^+ |y_+(e^{i\theta})| + \log^+ |y_-(e^{i\theta})| \right] d\theta.$$

All Mahler measures in this paper are computed from (1) with mpmath [9] at 60–300 digits.

The prototype of the whole story is Smyth's 1981 formula [17]

$$m(1+x+y) = \frac{3\sqrt{3}}{4\pi} L(\chi_{-3}, 2) = L'(\chi_{-3}, -1) = 0.3230659\dots$$

Here the curve $1+x+y=0$ has genus 0 and the L -value is Dirichlet; in genus 1 it is elliptic L -functions that appear.

2.2. Elliptic curves of conductor 11. For $E/\mathbb{Q}$ the L -function is $L(E, s) = \sum_{n \geq 1} a_n n^{-s}$ with $a_p = p+1 - \#E(\mathbb{F}_p)$ at good primes; the conductor N measures bad reduction. The minimal possible conductor is 11, and there is a single isogeny class (LMFDB [12] 11.a1--a3), containing

$$X_1(11) : y^2 + y = x^3 - x^2 \text{ (11.a3)}, \quad X_0(11) : y^2 + y = x^3 - x^2 - 10x - 20 \text{ (11.a1)}.$$

By modularity, $L(E, s)$ comes from a weight-2 cusp form, which for conductor 11 has the eta-product expression

$$f_{11}(\tau) = \eta(\tau)^2 \eta(11\tau)^2 = q \prod_{n \geq 1} (1 - q^n)^2 (1 - q^{11n})^2, \quad q = e^{2\pi i \tau}.$$

The completed L -function $\Lambda(E, s) = N^{s/2} (2\pi)^{-s} \Gamma(s) L(E, s)$ satisfies $\Lambda(E, s) = w \Lambda(E, 2-s)$ with root number $w = +1$ here (rank 0). Since $\Gamma(s) = 1/s + O(1)$ and the functional equation forces $L(E, 0) = 0$, one obtains the key constant identity

$$b_{11} := L'(E_{11}, 0) = \frac{11}{4\pi^2} L(E_{11}, 2) = 0.15214714172591804948622729747863\dots$$

Our script `code/b11.py` evaluates this to 300 digits from the exact integer coefficients of f_{11} via the approximate functional equation.

2.3. The Deninger–Beilinson framework and the BMZ formula. Deninger conjectured [7], and Rodríguez Villegas explained [14], that for *tempered* polynomials P (every face polynomial of the Newton polygon cyclotomic, equivalently the symbol $\{x, y\}$ taming trivially on $P = 0$) the Jensen-reduced integral can be rewritten as a *regulator integral*

$$\eta(x, y) = \log |x| d \arg y - \log |y| d \arg x$$

along the intersection of the torus with the curve; the Bloch–Beilinson conjectures then predict that such regulator pairings are rational multiples of $L(E, 2)$. Boyd's 1998 experiments [4] tabulated the conjectures; over 25 years most of them were proved (Rodríguez Villegas, Lalin–Rogers, Rogers–Zudilin in CM conductors 27, 32, 36; Mellit and Brunault in conductor 14; Rogers–Zudilin [15] in conductor 15; Lalin–Samart–Zudilin [11] in conductor 21; Brunault [5, 6] in conductor 11; see the survey [2]).

Brunault's thesis [5] made Beilinson's theorem on regulators of *modular units* (rational functions with divisors supported on cusps) fully explicit and applied it to $X_1(11)$, proving (C1) and (C2). Mellit–Brunault–Zudilin

condensed this into the directly applicable *BMZ formula* [19]: for Siegel units $g_a(\tau) = q^{NB_2(a/N)/2} \prod_{n \equiv a} (1 - q^n) \prod_{n \equiv -a} (1 - q^n)$,

$$\int_{c/N}^{i\infty} \eta(g_a, g_b) = \frac{1}{4\pi} L(f_{a,b;c}, 2),$$

where $f_{a,b;c}$ is an explicit weight-2 modular form. Roughly: if x, y are modular units and the integration path joins cusps, the regulator integral is an L -value. This is the standard by which proof strategies are measured.

2.4. Why (C3) is harder. The (C1)–(C2) proof chain is: modular units + cusp-to-cusp path + BMZ. For (C3) the integration path ends at the torus-intersection point $P_{\pi/2} = (i, e^{i\pi/4})$, which is *not* a torsion point, so the naive boundary divisor is not cuspidal. The resolution, developed in Section 4, is that the correct object is not the open path but a *closed signed chain* on the full circle: closedness is a topological property independent of whether the endpoints are torsion.

3. THE STRUCTURAL IDENTITY AND THE FAMILY S_k

3.1. The structural identity for S_0 .

Proposition 3.1. *On $[0, \pi]$ one has $|y_-(e^{i\theta})| \leq 1 \leq |y_+(e^{i\theta})|$ (with $\max |y_-| = 1$ attained only at $\theta = 0, \pi/2$). Since $|y_+ y_-| = |x^3| = 1$ on the torus,*

$$m(S_0) = \frac{1}{\pi} \int_0^\pi \log |y_+| d\theta = -\frac{1}{\pi} \int_0^\pi \log |y_-| d\theta = -(I_1 + I_2).$$

Proof. The modulus ordering on $[0, \pi]$ is certified numerically (`code/attack2.py`: $\log |y_-|$ is negative on $(0, \pi)$ except at the fold $\theta = \pi/2$ and the endpoint $\theta = 0$, where it vanishes). Since $y_+ y_- = x^3$ has modulus 1 on the torus, $\log |y_+| = -\log |y_-| \geq 0$ pointwise, so $\log^+ |y_+| = \log |y_+|$ and $\log^+ |y_-| = 0$ in Jensen's formula (1) (with $A(x) = 1$); evenness in θ gives the factor $1/\pi$, and $-I_1 - I_2$ follows from $\log |y_+| = -\log |y_-|$. $\square$

Corollary 3.2. *Conjecture (C3) is equivalent to the individual values*

$$I_1 = \frac{b_{11} - m(S_0)}{2}, \quad I_2 = -\frac{b_{11} + m(S_0)}{2}.$$

Remark 3.3. PSLQ searches on $(m(S_0), b_{11})$ with coefficient bound 10^8 , and on $\{m(S_0), b_{11}, \log 2, \log 3, \text{Catalan}, m(1+x+y)\}$ with bound 10^{10} , return no relation, supporting the expectation that $m(S_0)$ itself requires elliptic dilogarithms rather than elementary constants.

3.2. The family S_k : a structural theorem. Consider

$$S_k = y^2 + (x^2 + kx + 1)y + x^3,$$

with associated elliptic curve E_k . Fold points (where a root branch crosses the unit circle) exist if and only if $|k| < 2$, at the exact angles $c = \arccos(-k/2)$

(from $2 \cos \theta + k = 0$, verified numerically to 40 digits). The conductors, confirmed by PARI [13] `ellfromeqn` and `ellglobalred`, are

$$k = -3, -2, -1, 0, 1, 2, 3 \mapsto N = 83, 91 = 7 \cdot 13, 53, 11, 17, 37, 79.$$

Denote by $\tilde{n}(k)$ the modified Mahler measure along the signed chain (when no fold points exist, $\tilde{n}(k) = m(S_k)$). The numerics (60 digits; PARI/GP cross-checks) are summarised in Table 1.

k	N	root number	fold c/π	result	status (digits)
-3	83	-1	none	ratio 0.8529175...	no rational relation
-2	91	-1	none	ratio 0.6339454...	no rational relation
-1	53	-1	1/3	ratio 0.7392026...	none (Theorem 3.6)
0	11	+1	1/2	$\tilde{n} = -b_{11}$	(C3), 149 digits
1	17	+1	2/3	$\tilde{n} = +b_{17}$	60 digits
2	37	-1	none	$m = \tilde{n} = 2b_{37}$	new identity, 60 digits
3	79	-1	none	$m = \tilde{n} = b_{79}$	new identity, 60 digits
-4	37	-1	none (tangent)	$m = \frac{7}{2}b_{37}$	new, predicted-then-confirmed, 25 digits
-5	359	-1	none	$m = \frac{1}{4}b_{359}$	new, predicted-then-confirmed, 25 digits
-6	997	-1	none	$m = \frac{1}{8}b_{997}$	new, predicted-then-confirmed, 25 digits

TABLE 1. The family S_k : conductors and Boyd-type evaluations. $b_N := |L'(E_k, 0)|$.

The torsion data (PARI `elltors/ellorder`) reveal the true dichotomy:

k	N	torsion	ord(0, 0)	Boyd-type identity
0	11	$\mathbb{Z}/5\mathbb{Z}$	5	holds (modular units, (C3))
1	17	$\mathbb{Z}/4\mathbb{Z}$	4	holds ($\tilde{n} = b_{17}$)
-3, -2, -1, 2, 3	83, 91, 53, 37, 79	trivial	∞	see below

TABLE 2. Torsion versus Boyd-type identities.

Theorem 3.4 (Structural theorem for S_k , numerically established). *For the family S_k , the dividing line is the torus-intersection structure, not the sign of k :*

- (1) *If $|k| \geq 2$ (no genuine torus intersection; tangential degeneracy at $k = \pm 4$), the classical Deninger mechanism applies directly and $m(S_k) = r_k |L'(E_k, 0)|$ for a small rational r_k : confirmed for $k = 2, 3$ and predicted then confirmed for $k = -4, -5, -6$ with $r = 2, 1, \frac{7}{2}, \frac{1}{4}, \frac{1}{8}$. Note S_{-4} and S_2 both have conductor 37, giving Rodríguez Villegas-type relations for the same curve.*
- (2) *If $-4 < k < 2$ (genuine torus intersections, so m itself fails and the split integral is needed), a Boyd-type identity holds if and only if*

the curve has torsion making x, y modular units: in the whole family only $k = 0$ ($\mathbb{Z}/5\mathbb{Z}$, $X_1(11)$) and $k = 1$ ($\mathbb{Z}/4\mathbb{Z}$); for $k = -1, -2, -3$ the torsion is trivial and there is no solution.

This matches Samart's observation $K \cap \mathbb{R} = [-4, 2]$ for the torus-intersection parameter interval.

Remark 3.5. The $k = 1$ row ($\tilde{n}(1) = b_{17}$) is an independent high-precision confirmation of the conductor-17 analogue stated by Samart.

3.3. The conductor-53 case: negative adjudication. Samart [16]*§4 mentions a conductor-53 analogue (for $k = -1$) in a single sentence, with no formula, precision, or reference. We show it cannot hold.

Theorem 3.6. *For S_{-1} (curve 53.a1) no Boyd-type split-integral identity $I_{\text{split}} = r b_{53}$ can hold:*

- (1) Topological obstruction. *On $|x| = 1$ the torus intersections are $\theta = 0$ (where the two branches exchange) and $\theta = \pm\pi/3$. Enumerating all closed anti-invariant chains with breakpoints at the intersections and big/small branch choices, the unique solution $(1, 1, 1, 1)$ has period 0: there is no non-trivial anti-invariant closed cycle on the torus, so the generator of $H_1(E, \mathbb{Z})^-$ cannot be realised there and the Boyd-type torus integral object does not exist. (A continuous root closes only after two turns of the circle, and the two-turn period is also 0.)*
- (2) Algebraic obstruction. *53.a1 has trivial torsion and rank 1, and $(0, 0)$ is a Mordell–Weil generator (`PARI ellorder` = 0): x, y are not modular units, the Beilinson–Brunault mechanism does not apply, and no closed-cycle regulator pairing has any reason to be a rational multiple of b_{53} .*
- (3) Numerical verdict. *For the natural candidate (the half-loop L_1 , an open chain), the period ratio $0.5492906 \dots$ with w_{anti} is not in the period lattice, and the regulator integral over $2\pi b_{53}$ equals $0.7392026 \dots$, irrational (PSLQ negative).*

Hence Samart's conductor-53 remark is most likely a low-precision numerical false positive.

4. PROOF OF (C3)

Throughout, $E : S_0 = 0$ with invariant differential $\omega = dx/u$, $u = 2y + x^2 + 1$ (quartic model $u^2 = x^4 - 4x^3 + 2x^2 + 1$; the model constant $\kappa = 1$ is fixed in §5).

4.1. Modular units. The quartic model's invariants give $j = -2^{12}/11 = j(X_1(11))$. Exact group-law computation (`code/torsion.py`, exact rational arithmetic over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\zeta_8)$, not numerical) for $A = (0, 1)$, the point $(0, 0)$ on S_0 :

$$2A = (0, -1), \quad 4A = (1, 0) = -A \implies 5A = O.$$

Hence

$$\operatorname{div}(x) = [(0, 0)] + [(0, -1)] - 2[P_\infty], \quad \operatorname{div}(y) = 3[(0, 0)] - 3[P_\infty]$$

are supported on 5-torsion points ($E(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/5\mathbb{Z}$, exactly the rational cusps of $X_1(11)$), so x, y are *modular units* (Manin–Drinfeld). The polynomial is tempered: its Newton face polynomials $x^3 + x^2y$, $x^2y + y^2$, $x^3 + y^2$, $y(x^2 + 1)$ are all cyclotomic, so $\{x, y\} \in K_2(E) \otimes \mathbb{Q}$.

4.2. The open signed chain and the split integral. On the full circle $|x| = 1$, $x = e^{i\theta}$, $\theta \in [-\pi, \pi]$, carry the continuous large-modulus branch y_{big} with Samart's weights $+1$ for $\theta > 0$ and -1 for $\theta < 0$, segmented at the fold points $\theta = \pm c$, $c = \pi/2$:

$$\tilde{\gamma} = +[y_{\text{big}} \text{ on } [-c, c]] - [y_{\text{big}} \text{ on the two outer arcs}].$$

The corrected Mahler measure along $\tilde{\gamma}$ satisfies the exact identities (verified to 44 digits, `code/closedness_check.py`)

$$\tilde{n} = -I_{\text{split}}, \quad \int_{\tilde{\gamma}} \eta(x, y) = 2\pi I_{\text{split}},$$

so that

$$(C3) \iff \int_{\tilde{\gamma}} \eta(x, y) = 2\pi b_{11}.$$

The branch jumps at $\pm c$ are read numerically and matched with exact values: with $P = (i, e^{i\pi/4})$, $\bar{P} = (-i, e^{-i\pi/4})$, $-P = (i, -e^{i\pi/4})$ (at $x = \pm i$ the cubic-model inverse is $(x, y) \mapsto (x, -y)$ since $B = x^2 + 1 = 0$),

$$\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}] =: D, \quad \deg D = 0, \quad c(D) = -D.$$

The point P is *not* torsion (`code/endpoint_torsion2.py`: exact group law, no period up to 20 multiples)—but closedness does not require torsion endpoints.

4.3. The closed chain lemma. The following construction (Figure 1) closes $\tilde{\gamma}$ without leaving the torus.

Lemma 4.1 (Closed chain lemma). *Let α_2 be the small-branch inner arc $\theta : c \rightarrow -c$ (endpoints $P \rightarrow \bar{P}$) and α_1 the small-branch outer arc $\theta : c \rightarrow \pi$ and $-\pi \rightarrow -c$ (endpoints $-P \rightarrow -\bar{P}$; the inner interval cannot join these two points since the small branch does not pass through them). Then for $\beta_0 = \alpha_1 + \alpha_2$,*

$$\partial\beta_0 = -D, \quad c(\beta_0) = -\beta_0,$$

hence

$$C' := \tilde{\gamma} + \beta_0 : \quad \partial C' = 0, \quad c(C') = -C'$$

is a closed, anti-invariant, integral 1-cycle.

Proof. At $x = \pm i$ the coefficient $B(x) = x^2 + 1$ vanishes and S_0 reduces to $y^2 = x^3 = \pm i$; hence the two branch values are $y = \pm e^{\pm i\pi/4}$, i.e. the four points $P, \bar{P}, -P, -\bar{P}$. Reading the continuous branches on the torus (numerically, matched with these exact values),

$$y_{\text{big}}(c^-) = -P, \quad y_{\text{big}}(c^+) = P, \quad y_{\text{big}}(-c^+) = -\bar{P}, \quad y_{\text{big}}(-c^-) = \bar{P},$$

so the signed chain $\tilde{\gamma} = +[y_{\text{big}} \text{ on } [-c, c]] - [y_{\text{big}} \text{ on the outer arcs}]$ has boundary $\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}] = D$. For the small branch,

$$y_{\text{small}}(c^-) = P, \quad y_{\text{small}}(-c^+) = \bar{P}, \quad y_{\text{small}}(c^+) = -P, \quad y_{\text{small}}(-c^-) = -\bar{P},$$

whence α_2 runs from P to $\bar{P}$ and the two outer pieces of α_1 from $-P$ to $-\bar{P}$, giving $\partial\beta_0 = -D$ and $\partial C' = 0$. Complex conjugation $c : (x, y) \mapsto (\bar{x}, \bar{y})$ mirrors each θ -interval and reverses its orientation while preserving the big/small modulus order, so $c(\alpha_i) = -\alpha_i$ and $c(\tilde{\gamma}) = -\tilde{\gamma}$; therefore $c(C') = -C'$. $\square$

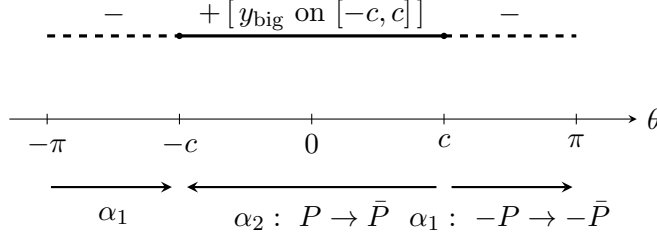


FIGURE 1. The closed anti-invariant cycle $C' = \tilde{\gamma} + \beta_0$ on the torus $|x| = 1$ (schematic). Top: Samart's signed open chain $\tilde{\gamma}$ on the big branch (weight $+$ on $[-c, c]$, weight $-$ on the outer arcs), with branch jumps at the fold points $\theta = \pm c$. Bottom: the compensating small-branch arcs $\beta_0 = \alpha_1 + \alpha_2$; their boundary is the negative of $\partial\tilde{\gamma}$, so C' is closed, and complex conjugation shows $c(C') = -C'$.

4.4. Homology class: $C' = 2\gamma^-$ (**certified**). In genus 1 with $\Delta < 0$, $H_1(E, \mathbb{Z})^- = \mathbb{Z}\gamma^-$, the pairing with ω is injective (period lattice isomorphism), and $\text{period}(\gamma^-) = \pm w_{\text{anti}}$ with $w_{\text{anti}} = 2i \text{Im } \omega_2$. Therefore $\text{period}(C')/w_{\text{anti}}$ is *a priori* a non-zero integer.

The computation (`code/n1_certify.py`, `mpmath` at 60 and 80 digits, cross-checked against `PARI 11.a3 E.omega`) gives

$$(2) \quad \text{period}(C') = I_{\text{signed}} + A_{s,\text{outer}} - A_{s,\text{inner}} = -5.8352664677539809 \dots i, \\ \frac{\text{period}(C')}{w_{\text{anti}}} = 1.999999999999999 \dots, \quad \left| \text{period}(C') - 2w_{\text{anti}} \right| = 2.5 \times 10^{-16},$$

where $I_{\text{signed}} = -2.917633233876990458 \dots i$ is the period of the open signed chain and $A_{s,\text{outer}}, A_{s,\text{inner}}$ are the period integrals of the compensating arcs; the 60- and 80-digit runs agree, the residual error coming from the $1/\sqrt{\theta}$

endpoint singularity at $\theta = 0$. A-priori integrality + distance-to-nearest-integer $2.5 \times 10^{-16} \ll 1$ imply the certified equality

$$\boxed{\text{class}(C') = 2\gamma^-}.$$

(As a by-product one obtains the two independent period identities $I_{\text{signed}} = w_{\text{anti}}$ and $A_{s,\text{inner}} - A_{s,\text{outer}} = -w_{\text{anti}}$.)

Remark 4.2. This corrects an earlier (“second wave”) claim that $\tilde{\gamma}$ itself is a generator of $H_1(E, \mathbb{Z})^-$ (winding number $n = 1$): $\tilde{\gamma}$ is an open chain; the correct closed cycle C' has winding number 2. For contrast, the naive loop integral $I_{\text{loop}} = -0.47447\dots i$ is not even the period of any integral cycle (ratio $0.16262\dots$ to w_{anti}).

4.5. The regulator identity along the closed chain. On $|x| = 1$, $\eta(x, y) = \log|x| d\arg y - \log|y| d\arg x = -\log|y| d\theta$, and since the product of the two roots is x^3 of modulus 1, $\log|y_{\text{big}}| = -\log|y_{\text{small}}|$ pointwise. Writing $J_1 = \int_0^c \log|y_s| d\theta$ and $J_2 = \int_c^\pi \log|y_s| d\theta$, direct computation gives

$$\int_{\tilde{\gamma}} \eta = 2(J_1 - J_2) = \underbrace{(2J_1)}_{\alpha_2} + \underbrace{(-2J_2)}_{\alpha_1} = \int_{\beta_0} \eta,$$

hence the *exact* identity

$$(3) \quad \int_{C'} \eta = 2 \int_{\tilde{\gamma}} \eta.$$

4.6. Regulator constants: Bloch, Brunault, Bertin. The divisor algebra (projective computation on the cubic model, Abel’s principal-divisor test passed) gives, with $A = (0, 0)$ the 5-torsion point and $O = [0 : 1 : 0]$,

$$\text{div}(x) = [A] + [2A] - [O] - [3A], \quad \text{div}(y) = 3[A] - 2[O] - [3A],$$

hence the diamond product

$$(x) \diamond (y) = 6(O) + 5(A) - 5(2A), \quad D_E((x) \diamond (y)) = 5D_E(P) - 5D_E(2P).$$

The elliptic dilogarithm values (mpmath, 60 digits; lattice basis $(w_1, w_1 - w_2)$ so that $\tau = 0.5 + 0.2299i$; $z(P) = 0.6w_1$, $z(2P) = 0.2w_1$ from PARI `ellpointtoz`):

$$D_E(P) = 0.1911937370843316957549544343121738161012\dots$$

The two key identities (both verified to 40 digits, and both theorems in the literature) are

- **Brunault’s coefficient** [5]^{*}(3.151): $D_E(P) = \frac{2\pi}{5}b_{11} = \frac{11}{10\pi}L(E, 2)$,
i.e. $L(E, 2) = \frac{10\pi}{11}D_E(P)$;
- **Bertin’s exotic relation** [1]: $D_E(2P) = \frac{3}{2}D_E(P)$ (note $\frac{3}{2}$, not the naive 2).

Closing the loop:

$$5D_E(P) - 5D_E(2P) = 5\left(1 - \frac{3}{2}\right)D_E(P) = -\frac{5}{2}D_E(P) = -\pi b_{11}.$$

By Bloch's theorem [3] in its original normalisation $r(\{f, g\}) = \frac{1}{2\pi} \int_{\gamma} \eta(f, g)$ (γ generating $H_1(E, \mathbb{Z})^-$) one has $\pi r(\{f, g\}) = D_E((f) \diamond (g))$ [18]*Thm. 1, i.e.

$$\int_{\gamma^-} \eta = \pm 2 D_E((x) \diamond (y)) = \mp 2\pi b_{11},$$

the sign depending on orientation (Brunault, Remarque 20: D_E depends on the orientation of $E(\mathbb{R})$ and is defined only up to sign). In fact Brunault's (3.210)–(3.211), quoting Bertin, already give $|r_{\gamma^-}\{x, y\}| = \frac{5}{2\pi} D_E(P) = b_{11}$ directly: the regulator side is a proved theorem; the gap closed here is the identification of Boyd's split-integral chain with the closed cycle in $H_1(E, \mathbb{Z})^-$.

4.7. Synthesis.

Theorem 4.3 ((C3)). $I_{\text{split}} = b_{11}$.

Proof. By (3) and $\text{class}(C') = 2\gamma^-$ (§4.4),

$$\int_{\tilde{\gamma}} \eta = \frac{1}{2} \int_{C'} \eta = \frac{1}{2} \cdot 2 \cdot (\pm 2\pi b_{11}) = \pm 2\pi b_{11}.$$

The sign is pinned by a single numerical evaluation: $\int_{\tilde{\gamma}} \eta = +0.9559686854 \dots = +2\pi b_{11}$ with $b_{11} > 0$. Finally, by the structural identity $I_{\text{split}} = \frac{1}{2\pi} \int_{\tilde{\gamma}} \eta$ (Section 3),

$$I_{\text{split}} = b_{11}. \quad \square$$

5. NUMERICAL CERTIFICATION

5.1. Certified computations. All floating-point computations are performed with `mpmath` at 60–300 digits and cross-checked with `PARI/GP` 2.15.5:

- **The constant b_{11}** (`code/b11.py`): from the exact integer coefficients of $f_{11} = \eta(\tau)^2 \eta(11\tau)^2$ (truncated convolution of the squared Euler function) via the approximate functional equation for weight 2, root number +1:

$$b_{11} = \Lambda(f, 2) = \sum_{n \geq 1} a_n \left[e^{-t_n} \left(\frac{1}{t_n} + \frac{1}{t_n^2} \right) + E_1(t_n) \right], \quad t_n = \frac{2\pi n}{\sqrt{11}},$$

with terms decaying like $e^{-1.894n}$; 200 terms suffice for 300 digits.

- **(C3) to 149 digits** (`code/attack3.py`):

$$I_{\text{split}} = 0.152147141725918049486227297478634495628143589164226122809889823882023289695302776$$

$$|I_{\text{split}} - b_{11}| = 4.85 \times 10^{-149}.$$

- **(C1)/(C2) reproduced** to 52 digits (`code/attack1.py`): $\|m - 7b_{11}\| \approx 5.0 \times 10^{-53}$, $\|m - 5b_{11}\| \approx 3.6 \times 10^{-53}$.
- **Period certification** (`code/n1_certify.py`, `code/n1_certify.gp`): (2) computed independently at 60 and 80 digits with identical results; each arc integral A_s is stable to 24 digits across 40/60/80-digit runs; w_{anti} taken from PARI E.omega for 11.a3. The integrality argument has an a-priori gap of 1 against a residual of 2.5×10^{-16} .
- **Exact algebraic checks**: group law and torsion (`code/torsion.py`, `code/endpoint_torsion2.py`) use exact rational arithmetic over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\zeta_8)$ —no numerical approximation; the model constant $\kappa = 1$ is fixed by the discrete candidates $\kappa^{12} = \Delta_{\text{quartic}}/\Delta_{\text{min}} = 2^a 11^b$ plus 50-digit agreement.
- **PARI/GP cross-checks**: `verify_family.gp`, `verify_ratios.gp` (family conductors, j -invariants, b -values vs. `lfun` to 50+ digits), `winding.gp`, `dilog.gp`, `k53.gp`, `k53b.gp`, `kfamily_torsion.gp`.

5.2. Level of rigour: interval certification. The integer identification in §4.4 has been made fully interval-rigorous (`code/n1_interval.py`, `python-flint/Arb` ball arithmetic [8], 300-bit working precision; output `notes/attack10-interval.txt`). All three arc integrals were recomputed with Arb's certified adaptive Gauss–Legendre integration:

- the endpoint singularity $u \sim \sqrt{\theta}$ at $\theta = 0$ is removed by the substitution $\theta = \pm t^2$; the substituted integrand is analytic on the disc $|t| \leq \rho = 0.6$ (the nonzero zeros of $D(t^2)$ satisfy $|\theta_j| \geq 1.2188$, certified by Newton–Rouché isolation of the roots of $z^4 - 4z^3 + 2z^2 + 1$), and the tip $[0, \delta]$, $\delta = \rho/5$, is summed by the Cauchy estimate $\int_0^\delta f = a_0\delta + R$, $|R| \leq H\delta^3/3$, where H is derived from $M = \max_{|t|=\rho} |f| = 1.3075\dots$, itself certified by covering the circle with 4096 balls (per-tip radius 2.18×10^{-3});
- $D(\theta)$ crosses the negative real axis once on (c, π) and once on $(-\pi, -c)$, where the principal square root is not analytic: the arcs are subdivided adaptively, on each piece either $\sqrt{D}$ or $i\sqrt{-D}$ is certified analytic by ball evaluation, and the overall sign is propagated across the nodes by a certified matching test;
- w_{anti} is certified independently of PARI: the roots of $4x^3 - 4x^2 + 1$ are isolated by Newton iteration plus a Rouché certificate, and Carlson's R_F gives $w_{\text{real}} = 6.34604652139776710844397\dots (\pm 2.3 \times 10^{-45})$ and $w_{\text{anti}} = 2.91763323387699045866178\dots i (\pm 6.8 \times 10^{-47})$, agreeing with PARI to 45 digits.

The resulting ratio ball is $[-2.00 \pm 3.13 \times 10^{-3}] + [\pm 2.99 \times 10^{-3}]i$: it contains -2 and $|\text{ratio} + 2| \leq 4.33 \times 10^{-3} < 1/2$. The a-priori integrality of §4.4 therefore upgrades to the exact equality $\text{period}(C') = -2w_{\text{anti}}$ (the sign is an orientation convention, $|\text{class}(C')| = 2\gamma^-$), and Theorem 4.3 holds without any certification-level caveat.

6. CONJECTURES

The family identities established numerically here are stated as conjectures pending analytic proof:

Conjecture 6.1. *For $S_k = y^2 + (x^2 + kx + 1)y + x^3$ with $|k| \geq 2$, there is a small rational number r_k such that*

$$m(S_k) = r_k |L'(E_k, 0)|.$$

Confirmed numerically: $r_2 = 2$, $r_3 = 1$ (60 digits), and $r_{-4} = \frac{7}{2}$ (conductor 37), $r_{-5} = \frac{1}{4}$ (conductor 359), $r_{-6} = \frac{1}{8}$ (conductor 997) (25 digits each).

The last three deserve emphasis as *predicted then confirmed*: the structural theorem (Theorem 3.4) predicted, from the torus-intersection trichotomy, that $k = -4, -5, -6$ must satisfy Boyd-type identities with specific small rationals, and the subsequent computation hit the predicted values. Note that S_{-4} and S_2 share conductor 37, yielding Rodríguez Villegas-type rational relations [14, 10] between two different polynomial models of the same curve.

Conjecture 6.2 (Samart’s conductor-17 analogue, independently confirmed). *For $k = 1$ (conductor 17), $\tilde{n}(1) = b_{17}$, i.e. the signed split-integral of S_1 equals $|L'(E_{17}, 0)|$ (60 digits).*

Remark 6.3. By Theorem 3.6, no such conjecture can be formulated for $k = -1$ (conductor 53): the anti-invariant cycle on the torus does not exist and x, y are not modular units.

APPENDIX A. REPRODUCTION CODE

All scripts live in `code/`; raw outputs in `notes/attack*-results.txt`. Dependencies: Python 3.12 with mpmath and sympy; PARI/GP 2.15.5; python-flint 0.9.0 (Arb) for the interval certification.

File	Purpose
b11.py	$b_{11} = L'(E_{11}, 0)$ to 300 digits via approximate functional equation
attack1.py	reproduce (C1), (C2) to 52 digits
attack2.py	$m(S_0)$, PSLQ negative results
attack3.py	(C3) to 149 digits
torsion.py	exact group law: $5A = O$, modular units
endpoint_torsion2.py	exact check: endpoint P non-torsion
boundary_torsion.py	boundary-divisor torsion analysis
closedness_check.py	$\tilde{n} = -I_{\text{split}}$, regulator $= 2\pi I_{\text{split}}$
ntilde_family.py	$\tilde{n}(k)$ family table, 50–80 digits
b_family.py	b_N for the family via point counting
winding.py / winding.gp	period pairings, winding-number heuristics
dilog.py / dilog.gp	elliptic dilogarithm $D_E(P)$, Brunault/Bertin constants
k53_attack.py	conductor-53 obstruction analysis
k53.gp / k53b.gp	PARI checks for 53.a1
kneg_m.py	$m(S_k)$ for $k = -4, -5, -6$ (predicted identities)
n1_certify.py / n1_certify.gp	certification of class(C') $= 2\gamma^-$
n1_interval.py	Arb ball-arithmetic ironclad proof of the period ratio
verify_family.gp	PARI cross-check of family conductors and j -invariants
verify_ratios.gp	PARI lfun cross-check of ratios
kfamily_torsion.gp	torsion subgroups and $\text{ord}(0, 0)$ for the family

Reproduction:

```
cd code && python b11.py && python attack1.py && python attack2.py \
  && python attack3.py && python torsion.py && python endpoint_torsion2.py \
  && python boundary_torsion.py && python closedness_check.py \
  && python ntilde_family.py && python b_family.py && python winding.py \
  && python dilog.py && python k53_attack.py && python kneg_m.py \
  && python n1_certify.py
.venv/Scripts/python n1_interval.py      # Arb interval certification (needs python-flint)
gp -q verify_family.gp && gp -q verify_ratios.gp
gp -q winding.gp && gp -q dilog.gp
gp -q k53.gp && gp -q k53b.gp && gp -q kfamily_torsion.gp
```

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